

NEXUS - Multi-Agent Personal Auto-Scheduling Platform

Web Dashboard (localhost:8000)

Overview

AI-Times

Mailman

Wallstreet

News

DocVault

WebSocket + REST

FastAPI Server (main.py)

REST API + WebSocket + Static Files

ORCHESTRATOR

Resource Monitor

CPU/RAM/Disk

Alarm System

Thresholds

LLM Semaphore

1 agent at a time

Agent Supervisor

Auto-restart x3

Digest Scheduler

APScheduler Cron

Subprocess per agent

AI-Times

YouTube AI

Video Digest

Mailman

Gmail Classification

7 Categories

Wallstreet Wolf

Stock Tracking

+ Commentary

News Briefer

Multi-Category

Summarization

DocVault

RAG Document

Q&A

Cron trigger

LLM Inference



YouTube Data API v3

YouTube Data API v3

Gmail API

OAuth 2.0

Yahoo Finance

yfinance

NewsAPI.org

Local Files

PDF / DOCX / TXT

Ollama / Qwen3

Local LLM (localhost:11434)

300s timeout | Semaphore(1)

SQLite

data/platform.db

6 Tables | Async (aiosqlite)

Email Service

SMTP (aiosmtplib)

HTML Digest Emails

Configuration

.env (API keys, schedules)

pydantic-settings

APScheduler CronTrigger

Configurable schedules via UI

Database Tables

cached_videos

email_records

stock_snapshots

news_articles

system_metrics

agent_logs

LEGEND



LLM Inference



DB Read/Write



External API Call



Email Digest